

Analysis of Pellet Cluster Distributions and Growth Probabilities in Mice

HT

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Abstract

This report outlines the steps taken to characterize feeding-cluster sizes in mice, fit an exponential decay to the cluster-size distribution, and estimate the probability that a cluster of size x continues growing to size $x+1$. Furthermore, we provide confidence intervals for these conditional probabilities using the Wilson (score) method. The integration of these methods yields a comprehensive view of how feeding events evolve in size and how frequently they terminate.

1 Introduction

In our feeding behavior study, we monitored the number of pellets consumed by mice in discrete *clusters*, where each cluster is defined as a sequence of pellets separated by inter-pellet intervals of less than 60 seconds. Once the interval exceeds 60 seconds, the previous cluster ends and a new one begins. This approach provides a quantitative measure of how feeding progresses in short bursts.

Two key objectives emerged:

1. **Cluster Size Distribution:** Understand how feeding events are distributed among various cluster sizes (e.g., 1 pellet, 2 pellets, ...).
2. **Probability of Growth:** Determine the probability $P(x+1 | x)$ that a cluster of size x (i.e. a sequence of x pellets so far) will continue to at least $x+1$ pellets, and quantify uncertainty in that probability via confidence intervals.

2 Methods

2.1 Data Acquisition and Preprocessing

- **Subjects:** We aggregated data from multiple mice (e.g. 11 animals), often combining both sexes under certain conditions.
- **Feeding Clusters:** For each mouse, we extracted timestamps of pellet retrieval. Any time gap under 60 seconds was considered part of the same *cluster*.
- **Cluster Sizes:** We thus assigned each sequence of contiguous pellets (separated by < 60 s) a *cluster size*, $s \in \{1, 2, 3, \dots\}$.

2.2 Cluster-Size Distribution and Exponential Decay Fit

We let X represent the cluster size, and we counted how many feeding events (clusters) fell into each size x . Denote by

$$n_x = \text{number of clusters of size } x.$$

After normalizing or averaging across all mice, we can write:

$$\text{count at size } x = n_x.$$

A typical empirical observation is that n_x might decrease as x grows. To test whether this decrease follows an exponential pattern, we fit a model of the form:

$$y(x) = a e^{-bx},$$

where:

$$y(x) \approx \text{expected (or average) count at cluster size } x,$$

and a and b are parameters. The fitting step:

$$\min_{a,b} \sum_{x=1}^{x_{\max}} (n_x - a e^{-bx})^2$$

is typically done via nonlinear least-squares routines. The result is an estimated $\hat{a}$, $\hat{b}$, and a measure of goodness-of-fit, often R^2 :

$$R^2 = 1 - \frac{\sum_x (n_x - \hat{a} e^{-\hat{b}x})^2}{\sum_x (n_x - \bar{n})^2}.$$

2.3 Conditional Probability of Growth to the Next Cluster Size

In addition to the distribution itself, we consider how likely a cluster of size x is to grow to size $x + 1$. Let

$$P(x + 1 \mid x) = \text{the probability that a cluster of size } x \text{ continues to } x + 1.$$

Empirically, we compute

$$\hat{p}_x = \frac{(\# \text{ of clusters of size } x \text{ that reached } x + 1)}{(\# \text{ of clusters of size } x)},$$

if the sample size for x is nonzero. Each $\hat{p}_x$ is essentially a *binomial proportion* with n_x trials and k_x successes (i.e. k_x of those n_x clusters indeed grew to size $x + 1$).

2.4 Wilson Confidence Intervals for $P(x + 1 \mid x)$

To address statistical uncertainty in $\hat{p}_x$, we compute 95% Wilson (score) confidence intervals. The Wilson interval for a binomial proportion $\hat{p} = k/n$ at confidence level $1 - \alpha$ can be expressed in multiple, but equivalent, ways. One common version is:

$$\text{CI}_{\text{Wilson}} = \frac{1}{1 + \frac{z^2}{n}} \left[\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1 - \hat{p})}{n} + \frac{z^2}{4n^2}} \right],$$

where z is the critical value from the standard normal distribution (e.g. $z = 1.96$ for 95% confidence), n is the number of trials (n_x), and $\hat{p}$ is the observed proportion (k_x/n_x). By applying this interval across $x = 1, \dots, x_{\max} - 1$, we obtain:

$$\hat{p}_x = P(x + 1 | x), \quad \text{CI}_x = [\text{CI}_\ell, \text{CI}_u],$$

for each cluster size x . This method is more stable than the simpler “Wald” interval ($\hat{p} \pm z\sqrt{\hat{p}(1 - \hat{p})/n}$) when sample sizes or proportions are small or near 0/1.

3 Results and Discussion

3.1 Cluster-Size Distributions and Exponential Fits

After combining data from all mice (or a subset, e.g. `order=1` and `Diet=NR`), we observed the following distribution of cluster sizes:

$$n_1 = 108.64, n_2 = 68.36, n_3 = 66.64, \dots n_{10} = 8.36,$$

(*example values shown*). Fitting $y = a e^{-bx}$ yielded parameters such as $(\hat{a}, \hat{b}) = (139.55, 0.28)$ with a high goodness-of-fit $R^2 \approx 0.97$. This indicates that the frequency of clusters decreases approximately exponentially with larger cluster sizes.

3.2 Probabilities of Growth and Wilson Intervals

Next, we computed $P(x + 1 | x)$, the probability that a size- x cluster grows to $x + 1$. Empirical results were as follows (95% Wilson CIs in parentheses):

$$\begin{aligned} P(2 | 1) &= 0.6290 \quad (0.6012, 0.6559), \\ P(3 | 2) &= 0.9774 \quad (0.9640, 0.9858), \\ P(4 | 3) &= 0.7902 \quad (0.7593, 0.8181), \\ &\vdots \\ P(10 | 9) &= 0.6316 \quad (0.0000, 0.0481). \end{aligned}$$

(Example numeric results are shown.) We see that $P(3 | 2)$ is very high, suggesting that once a cluster reaches size 2, it almost always proceeds to size 3. By contrast, certain transitions like $P(10 | 9)$ have fewer total observations, leading to a wide confidence interval that indicates high uncertainty.

3.3 Interpretation

- **Cluster Decay vs. Growth Probability.** The exponential decay in the frequency of larger clusters suggests that feeding events generally terminate before reaching high pellet counts. However, the conditional probabilities for smaller sizes (e.g. $2 \rightarrow 3$, $3 \rightarrow 4$) can be quite high, meaning once a mouse starts forming a larger cluster, it often proceeds a few additional pellets before stopping.
- **Confidence Intervals.** The Wilson intervals highlight how reliably we can estimate each $P(x + 1 | x)$. Smaller or more extreme proportions (close to 0 or 1) with relatively low sample size n_x will yield broader intervals, reflecting greater uncertainty about that probability.

4 Conclusions

In summary, our analysis proceeds in four key steps:

1. **Cluster Counting and Distribution.** We identify how many feeding events belong to each size x and confirm that the distribution typically decreases with x .
2. **Exponential Decay Fitting.** We fit n_x to a model $a e^{-bx}$, obtaining parameters that quantify how quickly clusters become rare as size increases.
3. **Growth Probabilities.** We calculate $\hat{p}_x = P(x + 1 \mid x)$ as a simple ratio of *successes* (clusters that grew) to the total number of size- x clusters.
4. **Wilson (Score) Confidence Intervals.** For each $\hat{p}_x$, we construct a 95% Wilson interval to account for the sampling variability of that probability estimate.

Overall, these steps shed light on how pellet-consumption clusters evolve in time. By examining both the global distribution (via exponential decay fits) and the local transitions between consecutive sizes (via $P(x + 1|x)$ with confidence intervals), we gain a detailed behavioral signature of how feeding clusters unfold or terminate.